Youssef Ahmed

236-862-6460 | yahmed06@student.ubc.ca | github.com/Youssef0192 | linkedin.com/in/youssef-ahmed-41ba0a436

TECHNICAL SKILLS

Languages:	Java, Python, SQL, R, C, C++
Workflow & Integration:	n8n workflow automation, REST API integration, Supabase, Twilio
Cloud & Data:	AWS, Databricks, Terraform (Infrastructure as Code), Docker, Linux
Tools & Practices:	Git, JUnit 5, UML, technical documentation, Java Swing, Microsoft Office

WORK EXPERIENCE

Software Developer (AI Analytics & DevOps), Seen Education | *Python, AWS, Terraform* Aug 2026 – Present

- Developed a Python platform that analyzes and benchmarks student performance across grades, schools, and national datasets, including automated scale score calculations and statistical comparisons.
- Automated client-ready school reports (charts, comparisons, executive summaries) for non-technical stakeholders.
- Provisioning AWS infrastructure with Terraform (Infrastructure as Code) to support a hosted deployment across schools throughout the Middle East, serving thousands of students.

Ontology Researcher & Information Architect, Roots Solutions & Automation May – Aug 2026

- Translated how organizations operate – their people, assets, documents, operations, events, commitments, and transactions – into a universal ontology that serves as a structured context layer for AI.
- Developed a framework with formal, reusable definitions enabling AI to understand organizational structure, relationships, and processes across different companies and industries.

PROJECTS

Valet Parking App (Academic), Software Construction | *Java, JUnit 5, n8n, Supabase* Jan – Apr 2026

- Developed a Java desktop application with a Swing GUI and console UI backed by the same model, supporting vehicle check-in, slot allocation, retrieval, and real-time wait time estimation.
- Achieved high JUnit 5 test coverage across model and persistence layers and documented the architecture in UML.
- Built an event-driven n8n workflow that updates vehicle records in Supabase as cars enter or leave.

HR Screening Automation (personal) | *n8n, Docker, APIs* May 2026

- Automated an HR intake process: the workflow monitors incoming emails for resumes, transforms them into structured candidate records, and screens applications against specified selection criteria.

TECHNICAL EXTRACURRICULAR ACTIVITIES

Rogers x Databricks Hackathon | UBC, Vancouver | *Databricks, PySpark, SQL, Python, Next.js, FastAPI* Sep 2026

- Co-built CrowdCost with a 5-person team: a Next.js/TypeScript and FastAPI console that flags transit surges at UBC, Waterfront, and Park Royal 1 hour ahead (82% caught in a backtest) and recommends moving spare buses.
- Built a 9-task Databricks pipeline in PySpark and SQL over 21.5M cell-tower pings and TransLink data that cleaned the data our surge model trained on, plus a LightGBM forecaster (MLflow) and a SciPy bus optimizer.
- Placed 4th after pitching a live demo to judges from TransLink, PwC, Databricks, Rogers, West Vancouver, and UBC.

T-CAIREM National Health Datathon | UBC, Vancouver | *Python, XGBoost* Oct 2026

- Modeled ICU-transfer risk from the first 8 hours of an admission with a 9-person team from pharmacy, medicine, and data science, using 22,000+ de-identified St. Michael's Hospital inpatient encounters.
- Co-built the XGBoost model (ROC-AUC 0.743) in a secure research environment, after the team widened scope to all general internal medicine patients because stroke and 24-hour subsets were too small.

Amazon Robotics Hackathon | UBC, Vancouver Sep 2026

- Placed 14th of 49 teams (team of 4) on a real-world problem set by Amazon Robotics engineers: routing warehouse robots that carry inventory pods to pick stations, scored on delivery speed across hidden test cases.
- Benchmarked multiple routing strategies in pure Python and shipped a congestion-aware Dijkstra planner that predicts when busy aisles clear, with greedy robot-to-pod assignment, batched pickups, and dock clearing.

Internship, Roots Solutions & Automation Jun 2022 – Jul 2024

- Visited industrial sites across Egypt and the UAE to document operational issues and planned automation.
- Researched multi-view 3D reconstruction methods for a proposed multi-camera rig that scans parts as they move.

EDUCATION

2nd Year Bachelor of Science, Computer Science, Statistics Combined Major (avg: 88.5%) Sep 2025 – May 2030 (Expected)
University of British Columbia, Vancouver, BC